Linear Graphs 2: Gradient from a Graph

Name _____ Date _____ No calculator Version 1

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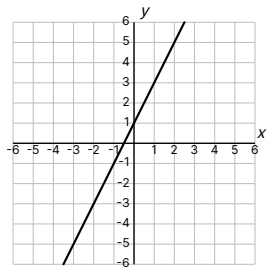
KEY WORDS gradient *(rōnaki)* · rise · run · rate *(pāpātanga)*

A Rise over run ●○○

Remember: Pick two points where the line crosses grid corners. Gradient = rise ÷ run. A line that goes down from left to right has a negative gradient.

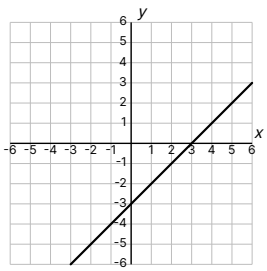
1. Find the gradient of each line.

a)



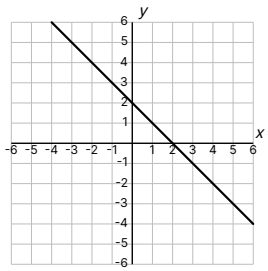
Gradient = _____

b)



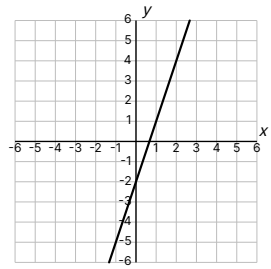
Gradient = _____

c)



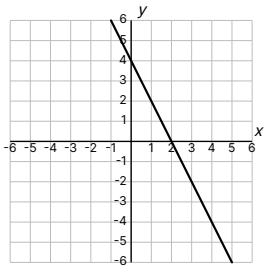
Gradient = _____

d)



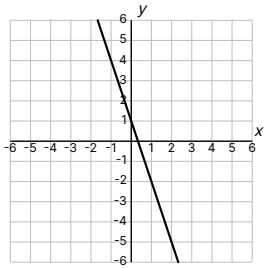
Gradient = _____

e)



Gradient = _____

f)



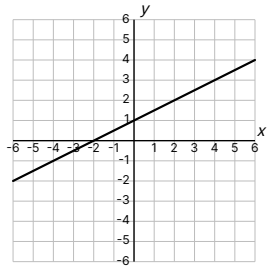
Gradient = _____

B Fractional gradients ●●○

Remember: If the line rises 1 for every 2 across, the gradient is $\frac{1}{2}$. Choose a run that makes the rise a whole number, and leave the answer as a fraction.

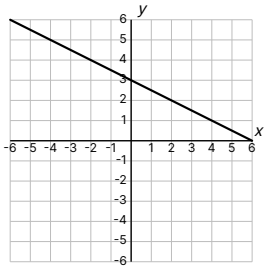
2. Find the gradient of each line. Give fractions in their simplest form.

a)



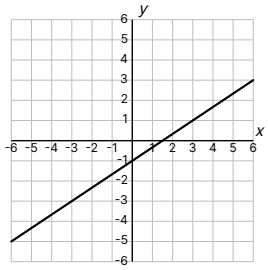
Gradient = _____

b)



Gradient = _____

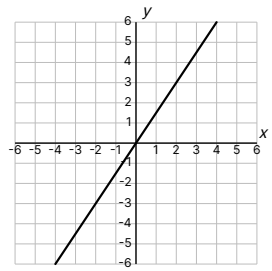
c)



Gradient = _____

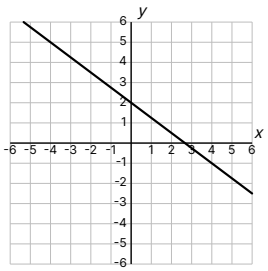
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d)



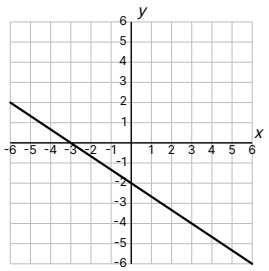
Gradient = _____

e)



Gradient = _____

f)

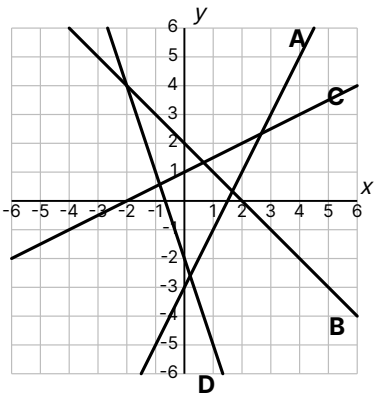


Gradient = _____

C Several lines on one grid ●●○

Remember: Each line is labelled where it leaves the grid. Follow the line back to find two grid corners it passes through.

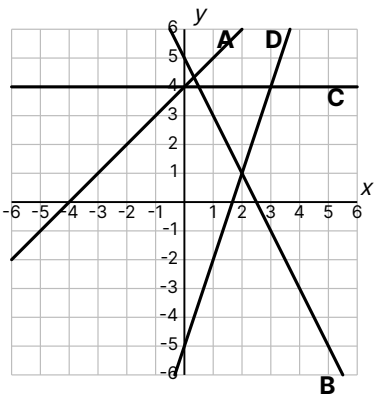
3. Find the gradient of each line on the grid.



a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____

4. Find the gradient of each line on the grid.



a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____

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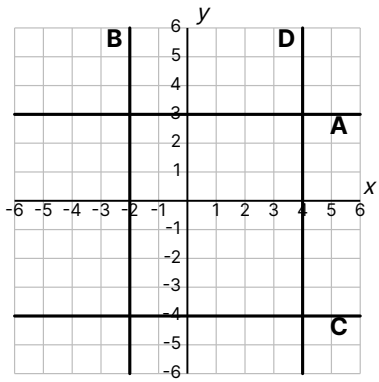
D Horizontal and vertical lines ●●○

Remember: A horizontal line has no rise, so its gradient is 0. A vertical line has no run, so its gradient is undefined. Every point on $y = 3$ has y-value 3; every point on $x = -2$ has x-value -2.

5. For each line on the grid, write down its gradient and its equation.

a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____



E Gradient from two points ●●○

Remember: You do not need the picture: gradient $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, the change in y divided by the change in x. Keep the points in the same order on the top and the bottom.

6. Find the gradient of the line through each pair of points. Give fractions in their simplest form.

a) $(1,\ 2)$ and $(3,\ 8)$ **b)** $(0,\ 5)$ and $(4,\ 1)$ **c)** $(-2,\ 1)$ and $(2,\ 3)$

d) $(2,\ -3)$ and $(5,\ 6)$ **e)** $(-1,\ 4)$ and $(3,\ -4)$ **f)** $(-3,\ -2)$ and $(1,\ 0)$

g) $(4,\ 7)$ and $(6,\ 2)$ **h)** $(-5,\ -1)$ and $(-2,\ 5)$ **i)** $(0,\ 0)$ and $(6,\ -4)$

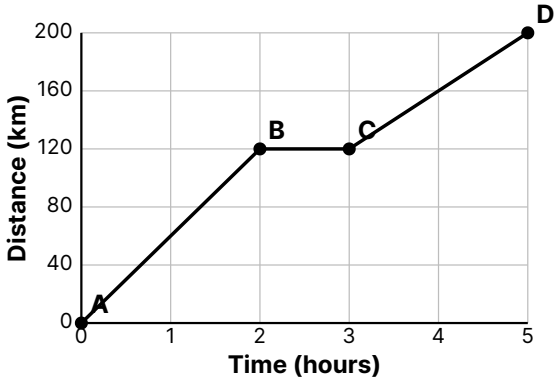
j) $(1,\ 1)$ and $(5,\ 7)$ **k)** $(-2,\ 3)$ and $(4,\ -6)$ **l)** $(0,\ -5)$ and $(5,\ 5)$

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F Gradient as a rate ●●●

Remember: On a distance-time graph the gradient is distance ÷ time, which is the speed. The steeper the line, the faster the travel. A flat part means no movement.

7. The graph shows Ana's journey. The gradient of each stage is a speed in km/h.



a) Gradient of stage AB

b) Gradient of stage BC

c) Gradient of stage CD

d) Which stage was the fastest? How can you tell from the graph?

e) What was happening during stage BC?

f) What was the average speed for the whole journey?

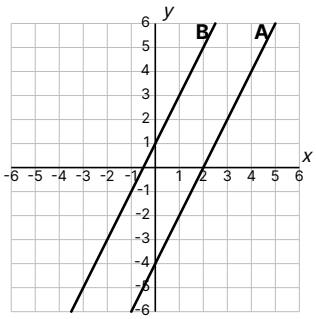
★ Extension

8. Find the gradient of the line through $(2,\ 7)$ and $(8,\ 3)$. Give it as a fraction in its simplest form.

9. Lines A and B are drawn on the grid.

a) Find the gradient of each line.

b) What do you notice about the two lines, and why does that happen?



10. A line passes through $(-2,\ 1)$ and $(4,\ 4)$.

a) Find its gradient.

b) Does the point $(10,\ 7)$ lie on the same line? Use gradients to explain.

11. A line has gradient $-\dfrac{3}{2}$ and passes through $(2,\ 5)$. Find the y-value of the point on the line where $x = 6$.

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Linear Graphs 2: Gradient from a Graph

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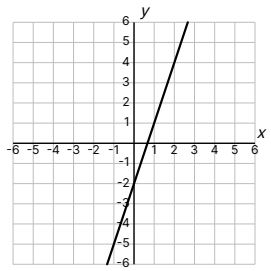
KEY WORDS gradient *(rōnaki)* · rise · run · rate *(pāpātanga)*

A Rise over run ●○○

Remember: Pick two points where the line crosses grid corners. Gradient = rise ÷ run. A line that goes down from left to right has a negative gradient.

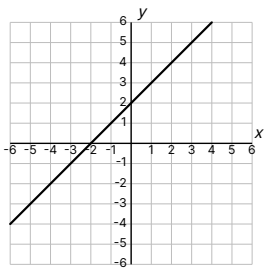
1. Find the gradient of each line.

a)



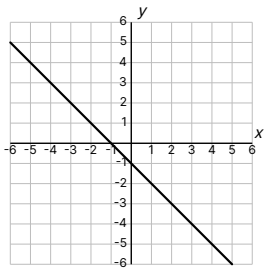
Gradient = _____

b)



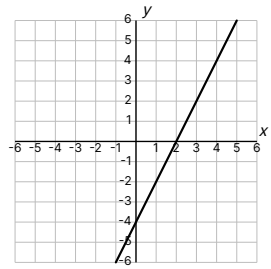
Gradient = _____

c)



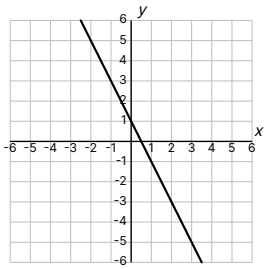
Gradient = _____

d)



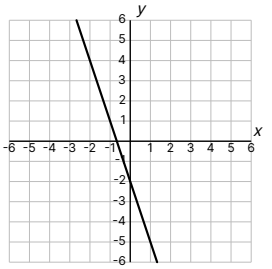
Gradient = _____

e)



Gradient = _____

f)



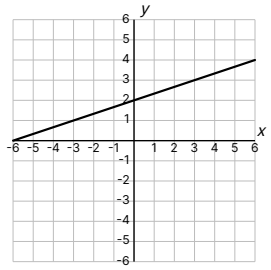
Gradient = _____

B Fractional gradients ●●○

Remember: If the line rises 1 for every 2 across, the gradient is $\frac{1}{2}$. Choose a run that makes the rise a whole number, and leave the answer as a fraction.

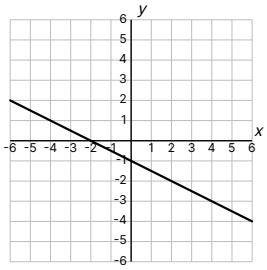
2. Find the gradient of each line. Give fractions in their simplest form.

a)



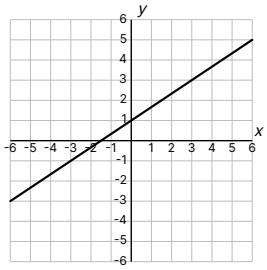
Gradient = _____

b)



Gradient = _____

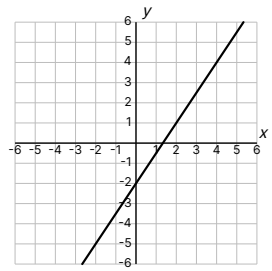
c)



Gradient = _____

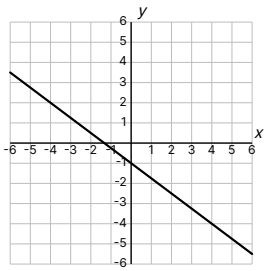
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d)



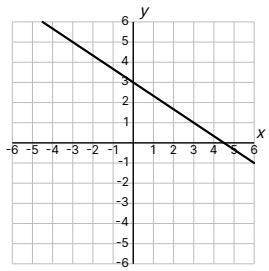
Gradient = _____

e)



Gradient = _____

f)

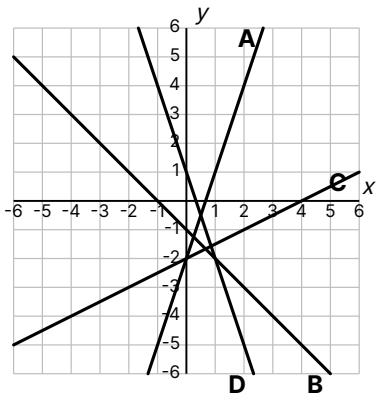


Gradient = _____

C Several lines on one grid ●●○

Remember: Each line is labelled where it leaves the grid. Follow the line back to find two grid corners it passes through.

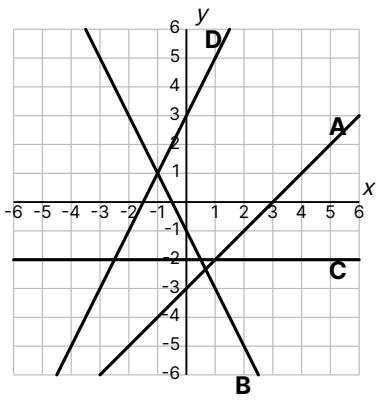
3. Find the gradient of each line on the grid.



a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____

4. Find the gradient of each line on the grid.



a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____

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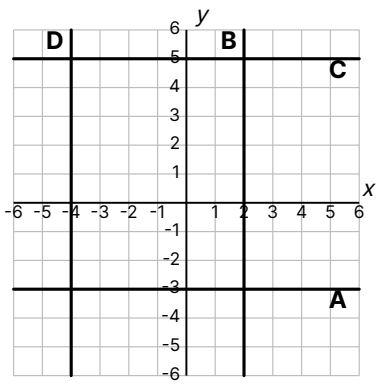
D Horizontal and vertical lines ●●○

> **Remember:** A horizontal line has no rise, so its gradient is 0. A vertical line has no run, so its gradient is undefined. Every point on $y = 3$ has y-value 3; every point on $x = -2$ has x-value -2.

5. For each line on the grid, write down its gradient and its equation.

a) Line A _____ **b)** Line B _____

c) Line C _____ **d)** Line D _____



E Gradient from two points ●●○

> **Remember:** You do not need the picture: gradient $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, the change in y divided by the change in x. Keep the points in the same order on the top and the bottom.

6. Find the gradient of the line through each pair of points. Give fractions in their simplest form.

a) $(2,\ 1)$ and $(4,\ 9)$ **b)** $(0,\ 6)$ and $(3,\ 0)$ **c)** $(-3,\ 2)$ and $(1,\ 4)$

d) $(1,\ -4)$ and $(3,\ 6)$ **e)** $(-2,\ 5)$ and $(2,\ -3)$ **f)** $(-4,\ -3)$ and $(2,\ 0)$

g) $(3,\ 8)$ and $(5,\ 3)$ **h)** $(-6,\ -2)$ and $(-3,\ 4)$ **i)** $(0,\ 0)$ and $(8,\ -6)$

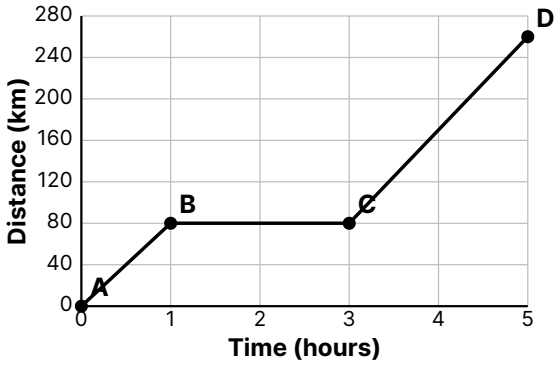
j) $(1,\ 2)$ and $(5,\ 8)$ **k)** $(-3,\ 4)$ and $(3,\ -5)$ **l)** $(0,\ -4)$ and $(4,\ 4)$

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F Gradient as a rate ●●●

Remember: On a distance-time graph the gradient is distance ÷ time, which is the speed. The steeper the line, the faster the travel. A flat part means no movement.

7. The graph shows Tom's journey. The gradient of each stage is a speed in km/h.



a) Gradient of stage AB

b) Gradient of stage BC

c) Gradient of stage CD

d) Which stage was the fastest? How can you tell from the graph?

e) What was happening during stage BC?

f) What was the average speed for the whole journey?

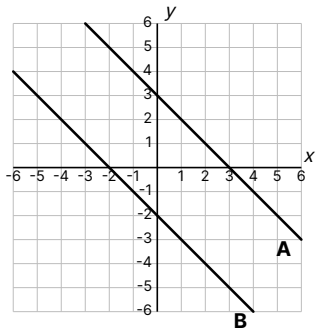
Extension

8. Find the gradient of the line through $(1,\ 8)$ and $(7,\ 4)$. Give it as a fraction in its simplest form.

9. Lines A and B are drawn on the grid.

a) Find the gradient of each line.

b) What do you notice about the two lines, and why does that happen?



10. A line passes through $(-3,\ -1)$ and $(3,\ 3)$.

a) Find its gradient.

b) Does the point $(9,\ 6)$ lie on the same line? Use gradients to explain.

11. A line has gradient $-\frac{2}{3}$ and passes through $(1,\ 4)$. Find the y-value of the point on the line where $x = 7$.

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